

Stephanie Vargas Aguilar, Ph.D.

Department of Molecular Biology · UT Southwestern Medical Center · Dallas, TX

stephanie.vargasaguilar@utsouthwestern.edu**EDUCATION**

- 2013 - 2018 **Ph.D., Biology and Immunology**
Humboldt University of Berlin, Berlin, Germany
Dissertation: "Investigation of the molecular mechanisms that enable the self-renewal of tissue-resident macrophages"
Mentor: Michael Sieweke, Ph.D.
- 2006 - 2012 **German Diploma (equivalent to B.A. + M.S.), Molecular Medicine**
Albert-Ludwigs-Universität Freiburg, Freiburg, Germany
Mentor: Rolf Kemler, Ph.D.

RESEARCH EXPERIENCE

- 2020–present **Postdoctoral Fellow**
UT Southwestern Medical Center, Dallas, TX
Mentor: Eric Olson, Ph.D.
- 2013–2018 **Graduate Student**
Centre d'Immunologie de Marseille-Luminy, Marseille, France
Max Delbrück Center for Molecular Medicine, Berlin, Germany
Mentor: Michael Sieweke, Ph.D.
- 2011–2012 **Research Technician**
Max Planck Institute for Immunobiology and Epigenetics, Freiburg, Germany
Laboratory of Rolf Kemler, Ph.D.
- 2008–2010 **Research Technician**
Albert-Ludwigs-Universität Freiburg, Institute for Molecular Medicine and Cell Research
Laboratory of Thomas Reinheckel, Ph.D.

PUBLICATIONS

1. **Vargas Aguilar S***, Dos Santos M, Tan W, Sanchez-Ortiz E, Chen K, Li H, Xu L, Liu N, and Olson EN*, A developmentally restricted $\gamma\delta$ T cell–IL-17A axis supports mouse heart regeneration, *Cell Reports*, in press. ***co-corresponding authors**
2. **Vargas Aguilar S**, Cui M, Tan W, Sanchez-Ortiz E, Bassel-Duby R, Liu N, and Olson EN. The PD-1–PD-L1 pathway maintains an immunosuppressive environment essential for neonatal heart regeneration. *Nature Cardiovascular Research*, March 2024. PMID: 38737787
3. **Vargas Aguilar S** (as part of the ImmGen Consortium). Network analysis of large-scale ImmGen and Tabula Muris datasets highlights metabolic diversity of tissue mononuclear phagocytes. *Cell Reports*, February 2023. PMID: 36708514
4. Subramanian S, Busch CJ, Molawi K, Geirsdottir L, Maurizio J, **Vargas Aguilar S**, Belahbib H, Gimenez G, Yuda RAA, Burkon M, Favret J, Gholamhosseinian Najjar S, de Laval B, Kandalla PK, Sarrazin S, Alexopoulou L, Sieweke MH. Long-term culture-expanded alveolar macrophages restore their full epigenetic identity after transfer in vivo. *Nature Immunology*, March 2022. PMID: 35210623
5. **Vargas Aguilar S** (as part of the ImmGen Consortium). ImmGen at 15. *Nature Immunology*, July 2020

6. Imperatore F, Maurizio J, **Vargas Aguilar S**, Busch CJ, Favret J, Kowenz-Leutz E, Cathou W, Gentek R, Perrin P, Leutz A, Berruyer C, Sieweke MH. SIRT1 regulates macrophage self-renewal. *EMBO Journal*, August 2017. PMID: 28701484
7. Matcovitch-Natan O, Winter DR, Giladi A, **Vargas Aguilar S**, Spinrad A, Sarrazin S, Ben-Yehuda H, David E, Zelada González F, Perrin P, et al. Microglia development follows a stepwise program to regulate brain homeostasis. *Science*, August 2016. PMID: 27338705
8. Dennemärker J, Lohmüller T, Müller S, **Vargas Aguilar S**, Tobin DJ, Peters C, Reinheckel T. Impaired turnover of autophagolysosomes in cathepsin L deficiency. *Biological Chemistry*, August 2010. PMID: 20536383

FUNDING

2025–present	Career Development Award PI \$231,000 American Heart Association
2022–2024	Postdoctoral Fellowship \$145,020 American Heart Association
2016–2017	Doctoral Scholarship €49,300 Foundation ARC pour la Recherche sur le Cancer — Aides doctorales en France
2013–2016	Doctoral Scholarship €79,457 International PhD Programme, Helmholtz Graduate School

MENTORSHIP EXPERIENCE

2023	Yu-Chung “Andy” Pien – Graduate Student
2017	Sethuraman Subramanian – Graduate Student

PRESENTATIONS

Oral Presentations

2024	II. Cardioimmunology Congress, Würzburg, Germany
2023	NPAW Symposium 2023, UT Southwestern Medical Center, Dallas, USA
2022	Charlas Virtuales CISEAL, Center for Research on Health in Latin America, Quito, Ecuador
2022	EMBO Workshop on Molecular and Cellular Basis of Regeneration and Tissue Repair, Barcelona, Spain
2020	Immunity in Regeneration Kick-off Mini Symposium, Center for Regenerative Therapies, Dresden, Germany
2019	33rd Annual Conference of the European Macrophage and Dendritic Cell Society, Marseille, France
2015	Scientific Retreat, Centre d'Immunologie Marseille-Luminy, Marseille, France

Poster Presentations

2023	BCVS Scientific Sessions, American Heart Association, Boston, USA
2022	NPAW Symposium 2022, UT Southwestern Medical Center, Dallas, USA
2018	EMBO Workshop on piRNAs and PIWI Proteins, Montpellier, France
2017	Nature Conference on Regeneration, Milan, Italy
2016	NeuroMac, Freiburg, Germany

2016 EMBO Workshop on piRNAs and PIWI Proteins, Montpellier, France

PROFESSIONAL SERVICE

2023–2024 **Vice President**

UT Southwestern Postdoctoral Association

2021–present **Chair of the Webinar Committee for “The Americas”**

International Society for Regenerative Biology

PROFESSIONAL MEMBERSHIPS

American Heart Association

International Society for Heart Research

International Society for Regenerative Biology

Latin American Society for Developmental Biology

Organization for Women in Science for the Developing World

SCIENTIFIC OUTREACH

2023 My personal and scientific path as a minority in STEM

Invited speaker, Hispanic Heritage Month Symposium, UT Southwestern Medical Center, Dallas, TX

2022 El anhelo de volver al corazón infantil

Invited speaker (virtual), Villaviciosa ConCiencia Science Outreach Association, Villaviciosa, Spain

Dallas, September 2026